

Cross-layer brokerage in area-crime-outcome police networks

ECMM447 CW2, UoE, 760012524

Abstract—Recorded crime data is typically published as separate frequency tables, hiding the relational structure that links areas, crime categories and recorded outcomes within every individual record. Six months of UK police-recorded crime for three forces are modelled as a weighted multilayer network with crime type as the shared mediator node between an area-crime layer and a crime-outcome layer. A four-component composite brokerage score (area spread, outcome connectivity, outcome entropy and weighted-distance betweenness) ranks crime categories as cross-layer brokers. Violence and sexual offences is the dominant broker in all three forces; drugs is a consistent second-tier broker not visible from raw event-count rankings. A PageRank robustness check gives similar broker rankings to betweenness (rank correlations 0.85 to 0.92) and a complementary cosine-similarity community detection on outcome profiles supports a broadly similar property-crime / violence-disorder community structure.

I. INTRODUCTION

Crime data is routinely published as separate frequency tables treating three connected dimensions as independent counts. Every crime record links an approximate area to a crime category, and a category to a recorded outcome. Categories that connect many local areas to many recorded outcomes are not visible from simple marginal rankings alone. Network analysis makes those connections the primary object of study rather than a side-effect of aggregation.

Generic graph datasets used in earlier module coursework lack the layered node-type structure needed to model area-crime-outcome relationships. I chose UK Police Open Data [1] because its three-field record structure (approximate location, crime category, recorded outcome) maps directly onto a multilayer bipartite design [2]. I excluded stop-and-search data because outcome availability differs across forces. I selected three England and Wales forces to allow regional comparison under a consistent legal and recording framework [3]. Open police data supports spatial analysis but has known accuracy, anonymisation and recording-completeness limits [3]. Prior work used bipartite projections to infer hidden offender ties [4]; this project instead models structural associations between areas, categories and outcomes, not person-level inference.

The research question is: *which recorded crime categories act as cross-layer brokers between Lower Layer Super Output Areas (LSOAs) and recorded police outcomes, and how consistent are these brokerage roles across Metropolitan Police Service, West Midlands Police and South Wales Police?*

II. DATA AND METHODS

A. Dataset and network construction

Street-level crime CSV files for November 2025 to April 2026 were downloaded from `data.police.uk` per force and reduced to force, LSOA code, crime type, and Last outcome category. Records without an LSOA were dropped from the area-crime layer and records without an outcome from the crime-outcome layer, keeping both layers per force to use the maximum available data. Table I summarises the resulting combined graphs. Missing-outcome rates differ substantially across forces (19.5% Metropolitan, 10.9% South Wales, 3.3% West Midlands); this is treated as a limitation because imputation would fabricate a mediator relationship the data does not support. Outcome label sets differ slightly across forces, so outcome connectivity is normalised within force and cross-force Spearman results indicate relative ranking stability, not equal outcome opportunity. Anti-social behaviour has no recorded outcome links (zero crime-outcome edge weight in all three forces) and is retained in the area-crime layer only.

TABLE I
DATA VOLUME AND COMBINED-GRAPH SIZE PER FORCE. ALL THREE FORCES SHARE 14 CRIME TYPE LABELS; AREA-CRIME LAYER HAS LSOAs PLUS CRIME TYPES, CRIME-OUTCOME LAYER HAS CRIME TYPES PLUS OUTCOME CATEGORIES, COMBINED GRAPH JOINS THEM THROUGH CRIME TYPE NODES.

Force	Records	Nodes	Edges
Metropolitan Police Service	540,963	7,142	56,343
West Midlands Police	152,098	1,776	17,398
South Wales Police	59,031	952	7,685

I chose the multilayer design over a single untyped graph because merging areas, crimes and outcomes into one node set would erase the mediator interpretation the brokerage question depends on. Each layer is an undirected bipartite graph, and the combined graph joins them through shared crime type nodes. Because area and outcome nodes connect only to crime nodes, crime types are the only class that can lie between an area and an outcome, so undirected betweenness captures the mediator role without a directed formulation. Edges are weighted by record count so that more frequent area or outcome associations contribute proportionally to path structure.

B. Cross-layer brokerage measurement

A single centrality measure would not distinguish a ubiquitous category from one that mediates between many areas and outcomes. The primary analysis ranks each outcome-linked

crime category on four structural dimensions within its force: the proportion of LSOAs where it appears, the proportion of outcome types it connects to, the Shannon entropy of its outcome distribution, and its weighted-distance betweenness [5] in the combined graph. Higher normalised entropy means a crime type’s outcomes are spread across several categories rather than one dominant outcome. Betweenness was approximated with $k=250$ and a fixed seed on the two larger graphs; South Wales was computed exactly. The four within-force percentile ranks are averaged into a single composite brokerage score, with equal weighting because the dimensions are not commensurable in raw scale. The score is exploratory; no full component-weight sensitivity analysis is included, so the equal-weight ranking should be interpreted as a transparent heuristic rather than a proven stable ordering.

Two Spearman checks were computed: brokerage score versus raw area-crime volume within each force (to test whether the composite reproduces frequency), and brokerage score between force pairs across shared categories (to test cross-force stability). At $n \approx 13$ per force both are descriptive rather than hypothesis tests. As a robustness check, PageRank [6] was computed on the combined graph and its ordering compared with betweenness. PageRank on an undirected weighted graph is not independent of degree, so agreement between the rankings supports rather than proves that the top brokers are not betweenness artefacts.

C. Outcome-profile similarity network

The similarity network is a pruned one-mode projection of the crime-outcome layer [6]. Each crime category is represented as a vector of within-crime outcome proportions and pairwise cosine similarity is computed; each node retains its three most similar neighbours to guarantee full participation. This top-3 rule is a descriptive pruning step for readability, not a statistical threshold. The resulting 13-node graph is partitioned by greedy modularity community detection [7], chosen over Louvain as NetworkX-native and interpretable at this scale. Community colours are semantic labels assigned to each greedy-modularity community from its crime-type members, not raw community IDs.

III. RESULTS AND DISCUSSION

Figure 1 shows the data overview. Force sizes differ substantially (Table I), so all subsequent comparisons use brokerage structure and within-force proportions rather than raw counts.

Figure 2 shows the brokerage rankings. Violence and sexual offences ranks first in all three forces (0.92 Metropolitan, 0.88 West Midlands, 0.86 South Wales) with betweenness concentrated at 0.87, 0.99 and 0.99 respectively. This concentration is consistent with a hub-like position in the combined crime network, though this project does not formally test a scale-free fit. Drugs ranks second in all three (0.66, 0.67, 0.74). A raw crime-frequency table for Metropolitan Police Service would not identify drugs strongly, because several categories have higher event counts. The composite ranks it second by combining geographic spread, outcome diversity and path

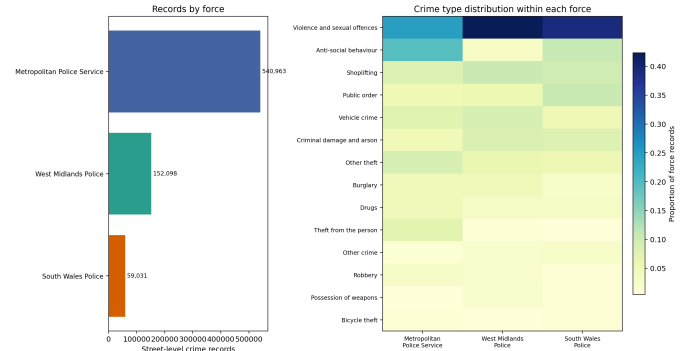


Fig. 1. Record volume and crime-type composition across the three forces. Metropolitan Police Service has more than nine times the records of South Wales; all cross-force comparisons use brokerage scores and proportions rather than raw counts.

position, properties a frequency table cannot capture at once, showing what the multilayer framing adds beyond raw counts. Second-tier patterns diverge after drugs. Other crime and other theft lead in Metropolitan, criminal damage and public order in West Midlands, public order in South Wales. These differences reflect measured variation in spread and outcome connectivity within the available recorded data.

As a descriptive consistency check, Spearman correlation between brokerage score and raw area-crime volume was moderate in Metropolitan ($\rho=0.57$) and West Midlands ($\rho=0.58$), and stronger in South Wales ($\rho=0.85$). The higher South Wales value may reflect smaller forces: with fewer LSOAs and less varied structure, spread and volume are harder to separate, so brokerage converges toward frequency. Cross-force brokerage correlations were high: 0.85 between Metropolitan and South Wales, 0.78 between Metropolitan and West Midlands, and 0.88 between South Wales and West Midlands, so broker role ordering is largely consistent across forces despite regional and scale differences. PageRank rankings on the combined graph agree closely with betweenness rankings (Spearman $\rho=0.85$ Metropolitan, 0.92 West Midlands, 0.89 South Wales), so the broker ordering does not depend on the betweenness definition alone.

Figure 3 shows the similarity networks. After relabelling raw modularity communities by crime-type membership, all three networks show a property-crime / violence-disorder split; West Midlands produces three raw modularity communities, which are collapsed into the broader semantic labels. The property-crime community groups burglary, vehicle crime, bicycle theft, other theft, shoplifting, and theft from the person, all sharing outcome profiles dominated by *investigation complete*; *no suspect identified* and *unable to prosecute suspect*. The violence-disorder community groups violence and sexual offences, drugs, public order, possession of weapons and other crime, with more distributed outcome profiles. Robbery shifts from the property community (Metropolitan, West Midlands) to the violence community (South Wales); the partition is otherwise broadly similar. Both the composite score and the

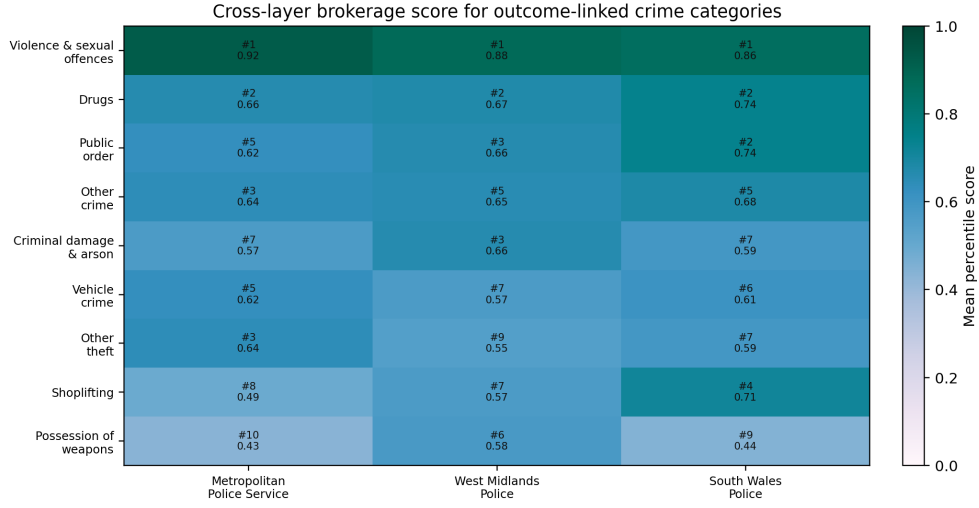


Fig. 2. Cross-layer brokerage scores for outcome-linked crime categories in each force. Violence and sexual offences ranks first in all three forces; drugs ranks second in all three despite not appearing in the Metropolitan top five by raw event volume. Anti-social behaviour is excluded because its crime-outcome edge weight is zero. (top brokers per force shown; equal scores share a rank).

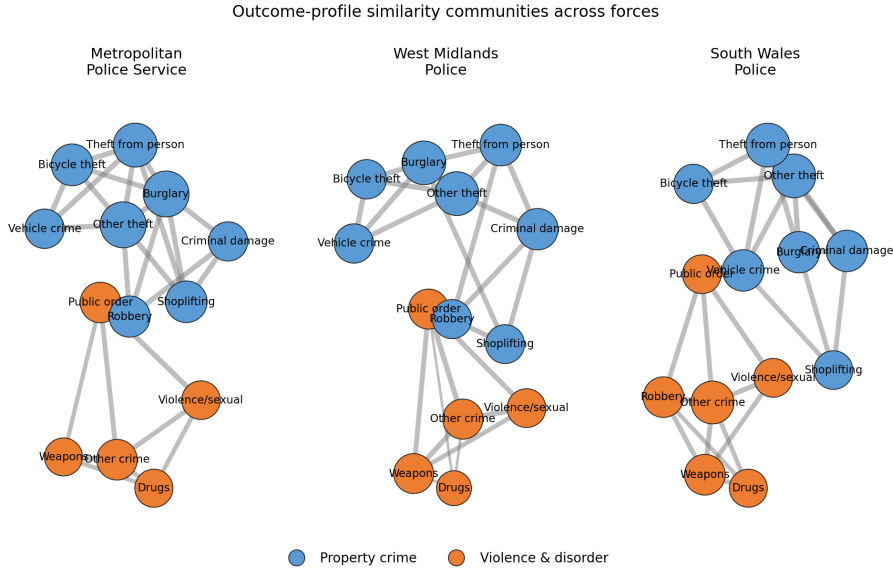


Fig. 3. Outcome-profile similarity networks per force (top-3 cosine-similarity edges per node). Node colour shows the greedy-modularity community relabelled by crime-type membership as Property crime or Violence & disorder, giving consistent colours across panels. After semantic relabelling of raw modularity communities, the panels show a broadly similar property / violence-disorder split; robbery shifts between them.

similarity communities place drugs in the diverse-outcome group, matching its elevated entropy.

Limitations: Betweenness is approximated ($k=250$, fixed seed) on the two larger graphs and, using inverse-weight distances, remains partly volume-sensitive. Area spread counts LSOA presence rather than population-adjusted risk, which may overstate reach in densely populated areas; LSOAs are approximate statistical areas, not exact locations. The six-month window captures one period only, so seasonal patterns are not visible. All data is police-recorded crime; no causal claims about crime, outcomes, or police performance are made.

IV. CONCLUSION

Violence and sexual offences is the dominant cross-layer broker in all three forces; drugs is a consistent second-tier broker not visible from raw event-count rankings alone. Broker ordering is broadly consistent across forces (Spearman 0.78 to 0.88); the same-graph PageRank comparison gives a similar ordering, suggesting the result is not purely a betweenness-ranking artefact, and the cosine-similarity community structure aligns with the composite ranking. The consistency suggests these structural roles may be partly crime-type related, although differences in recording practices and missing outcome fields cannot be ruled out.

Centrality-metric comparison

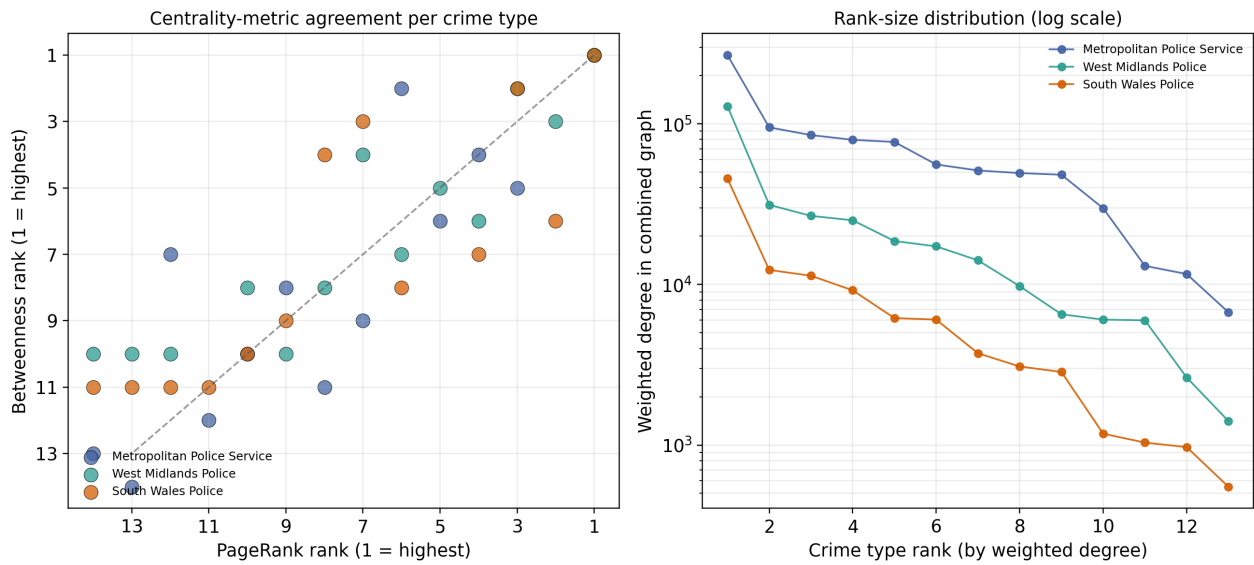


Fig. 4. Centrality-metric comparison. Left: PageRank rank vs betweenness rank per crime type; points near the dashed diagonal indicate the two metrics agree. Right: weighted-degree rank-size distribution of crime nodes in the combined graph, log-scaled; the steep drop from rank 1 is consistent with a highly concentrated weighted-degree distribution.

Figure 4 shows two supporting views of the combined crime network. Left: PageRank rank against betweenness rank per brokerage-eligible category; points near the diagonal indicate agreement, and violence and sexual offences sits at rank 1 for both metrics in all three forces. Right: rank-size distribution of weighted node degree on a log scale; the steep drop from the top-ranked category is consistent with a highly concentrated weighted-degree distribution without formally testing a scale-free fit.

Future work

Longer time windows would allow temporal-stability tests using leave-one-month-out resampling of the brokerage hierarchy. Area spread should be normalised by LSOA population to remove the implicit weighting toward densely populated areas. Additional forces would test whether the property-crime / violence-disorder community structure generalises beyond England and Wales, and inclusion of stop-and-search data, once outcome comparability is resolved, would extend the crime-outcome layer.

REFERENCES

- [1] UK Police Open Data, "Data downloads and documentation," 2026, accessed 23 June 2026. [Online]. Available: <https://data.police.uk/data/>
- [2] M. Kivelä, A. Arenas, M. Barthélemy, J. P. Gleeson, Y. Moreno, and M. A. Porter, "Multilayer networks," *Journal of Complex Networks*, vol. 2, no. 3, pp. 203–271, 2014.
- [3] L. Tompson, S. Johnson, M. Ashby, C. Perkins, and P. Edwards, "Uk open source crime data: accuracy and possibilities for research," *Cartography and Geographic Information Science*, vol. 42, no. 2, pp. 97–111, 2015.
- [4] H. Isah, D. Neagu, and P. Trundle, "Bipartite network model for inferring hidden ties in crime data," in *Proceedings of the 2015 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*. ACM, 2015, pp. 994–1001.
- [5] U. Brandes, "A faster algorithm for betweenness centrality," *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001.
- [6] M. E. J. Newman, *Networks: An Introduction*. Oxford University Press, 2010.
- [7] A. Clauset, M. E. J. Newman, and C. Moore, "Finding community structure in very large networks," *Physical Review E*, vol. 70, p. 066111, 2004.